



Listed below are some of the key aspects of green tech manufacturing.

1. Sustainability: The effort here is to convert waste into energy and make it a sustainable process. This helps in creating better societies that are cleaner as well, and lowers the stress on natural resources.

2. Pollution reduction: The idea is to reduce pollution in all formats, and to see how we can create a greener future by converting pollutants into energy -- this requires a change in production parameters and in thinking.

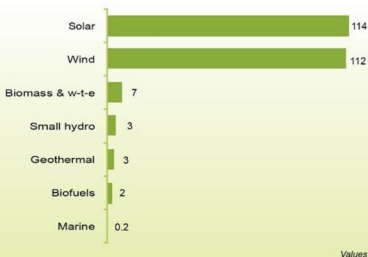
3. Provation theory: Here we must consider the built but obsolete assets that we have and try to repurpose them with innovation and creativity to align with current requirements. We must look at these assets and try to provate on them with the latest

technology. (Provate is a term derived from 'product' and 'innovate'.)

4. Framework-to-framework design: This is about making reusability the central requirement at the product creation phase itself.

5. Versatile manufacturing processes: Processes must be designed such that they can be aligned to serve multiple purposes, just by changing certain specs in the system.

The hot sectors in the green technology domain are energy, green nanotechnology and green chemistry. And these have enormous potential if the business models are aligned in the right sequence -- from managing the input costs to creating value and thereby delivering profits. Figure 2 shows global green investments for 2016.



Values are in US\$ billion

Fig. 2: Global green investments for 2016

Source: UNEP; Bloomberg New Energy Finance

Different aspects of green technology

In general, green tech initiatives involve environment-friendly methods of production, consumption and disposal. Therefore, investors and evangelists need to look at solutions ranging from product substitution to redesigning production facilities with newer technologies, and in some cases, they may have to completely overhaul factories.

So, green tech products should involve less wastage during production, less dependence on fossil fuels, and reusability of existing infrastructure to include the latest technological advances. Some of the major product categories to invest in are energy generation, green chemicals, sustainable or recyclable products, and alternate energy producing technologies like solar panels and thermal heating discs.

Apart from solar cells, other products/equipment worth investing in are rainwater harvesting systems (which are mechanical in nature and extremely inexpensive to set up), heating and insulation facilities, wind generation facilities, etc.

Challenges to green tech adoption

The primary challenge that organisations and corporations face in the adoption of green tech is that the initial investment is very high and, in most cases, the overall technology implementation cost is much higher than the technology it replaces. The real deterrent in this case is the cost of R&D, as green technology is in its nascent stages. The other barriers or challenges are regulatory in nature -- there is no set framework in terms of implementation nor is there a policy push from the states' end. So, for an investor this is the right time not only to enter the segment but also influence policy formulation within the country and globally. The positives include the development of a talent pool for this new sector, which with

